EXPERIMENT 00

“ALREADY INSTALLED ON MY SYSTEM THAT’S WHY I DON’T HAVE SCREENSHOTS”

WSL

Installing and Enabling WSL (Windows Subsystem for Linux)

WSL (Windows Subsystem for Linux) allows you to run a Linux distribution on Windows without needing a virtual machine or dual boot. Below are the steps to install and configure it properly.

Step 1: Install WSL and Ubuntu Distribution

Open **PowerShell as Administrator** and run:

wsl --install -d Ubuntu

This command:

- Installs WSL and enables required features (VirtualMachinePlatform and Windows Subsystem for Linux).
- Installs the **Ubuntu** distribution by default.

Note*: If you want another distribution, replace Ubuntu with the desired distro name.*

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Step 2: Verify Installation

Once the installation is complete, restart your computer and verify WSL is installed:

wsl --list --verbose

OR

wsl -l -v

This will show:

- Installed distributions
- Their versions (WSL 1 or WSL 2)
- Running status

If Ubuntu is installed but not set as the default, you can check the default distro:

wsl --list --verbose

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Step 3: Set a Default Distribution

To set Ubuntu as the default Linux distribution:

wsl --set-default Ubuntu

OR

wsl -s Ubuntu

Now, whenever you run wsl in the terminal, it will launch Ubuntu by default.

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Step 4: Check Installed Distributions

To view all installed Linux distributions:

wsl --list --all

If you want to install a new one, use:

wsl --install -d <DistroName>

For example:

wsl --install -d Debian

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Step 5: Check and Change WSL Version

WSL supports **WSL 1** and **WSL 2**. To check which version your distro is using:

wsl --list --verbose

To set a distribution to WSL 2:

wsl --set-version Ubuntu 2

To set a distribution to WSL 1:

wsl --set-version Ubuntu 1

To make **WSL 2** the default for all future installations:

wsl --set-default-version 2

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Fixing Common Errors

If you face errors during installation, try these fixes:

1. "WSL is not recognized as a command"

- Ensure you're running PowerShell as **Administrator**.
- Enable WSL manually:

dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart

Enable **Virtual Machine Platform** (needed for WSL 2):

dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart

Restart the system and try again.

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2. "WSL 2 requires an update to its kernel"

If you get this error when switching to WSL 2, update the kernel manually: 1. Download and install the latest **WSL 2 kernel update** from Microsoft's official page. 2. Try running:

wsl --set-version Ubuntu 2

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3. "Error: 0x80370102 – The virtual machine could not be started"

This happens if **Virtualization** is disabled in BIOS.

Fix:

- Restart your PC and enter **BIOS/UEFI** (Press F2, F10, Delete, or Esc depending on your PC).
- Look for **Intel VT-x / AMD-V** and **Enable** it.
- Save changes and restart.

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4. "DNS Resolution Issues Inside WSL"

If WSL is unable to connect to the internet:

echo "[network]" | sudo tee -a /etc/wsl.conf

echo "generateResolvConf = false" | sudo tee -a /etc/wsl.conf

sudo rm /etc/resolv.conf

echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf

Then restart WSL:

wsl --shutdown

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Uninstalling WSL

To completely remove WSL:

wsl --unregister Ubuntu

To disable WSL:

dism.exe /online /disable-feature /featurename:Microsoft-Windows-Subsystem-Linux

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Final Notes

- **wsl -d <DistroName>** runs a specific distro temporarily.

- **wsl --terminate <DistroName>** stops a running distro.
- **wsl --shutdown** stops all WSL instances.

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Now we will install git, docker, podman, virtualbox and vagrant in wsl

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Step 1: Install Git

```
sudo apt install git -y
git --version
```

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Step 2: Install Docker Engine

```
sudo apt install ca-certificates curl gnupg -y
sudo install -m 0755 -d /etc/apt/keyrings

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io -y
```

Verify:

```
docker --version
```

(Optional)

```
sudo usermod -aG docker $USER
```

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Step 3 (Alternative): Install Podman

sudo apt install podman -y

podman --version

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Step 4: Install VirtualBox (Windows Host)

Download and install:

https://www.virtualbox.org/wiki/Downloads

Verify:

VBoxManage --version

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Step 5: Install Vagrant

Download:

https://developer.hashicorp.com/vagrant/downloads

Verify:

vagrant --version

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Result

WSL, Ubuntu, Git, Docker/Podman, VirtualBox, and Vagrant were successfully installed.

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Read More

- WSL Documentation: https://learn.microsoft.com/windows/wsl/
- Docker Install Guide: https://docs.docker.com/engine/install/
- Podman Docs: https://podman.io/docs
- VirtualBox Docs: https://www.virtualbox.org/manual/
- Vagrant Docs: https://developer.hashicorp.com/vagrant/docs